

House Price Prediction Using Machine Learning

Project Summary

Project Objective: The objective of this project was to develop a machine learning model capable of predicting house prices using various housing and demographic features. Accurate house price prediction helps real estate businesses, investors, and buyers make informed decisions based on data-driven insights.

Dataset Overview: The California Housing Dataset was used for this analysis. The dataset contains information such as geographical location, housing age, number of rooms, population, household count, median income, and proximity to the ocean. The target variable for prediction was *median_house_value*.

Data Preprocessing: The dataset was explored to understand its structure, identify missing values, and examine feature distributions. Missing values in the *total_bedrooms* column were replaced using median imputation. Duplicate records were removed, and the categorical feature *ocean_proximity* was transformed into numerical variables using one-hot encoding to make the data suitable for machine learning models.

Exploratory Data Analysis: Statistical summaries, correlation analysis, and visualizations were performed to identify important patterns. Histograms, heatmaps, scatter plots, boxplots, and feature importance charts were used to understand relationships among variables. The analysis revealed that median income and geographical location were the strongest factors influencing house prices.

Model Development and Evaluation: Two machine learning algorithms were implemented: Linear Regression and Random Forest Regressor. The dataset was divided into training and testing sets, and model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Cross-validation was also applied to ensure model reliability and stability.

Results and Conclusion: The Random Forest Regressor outperformed Linear Regression by achieving higher predictive accuracy and lower error values. The study concluded that income levels, location, and housing characteristics significantly impact property values. The developed model demonstrates the effectiveness of machine learning techniques in solving real-world house price prediction problems and can support data-driven decision-making within the real estate industry.